

Supplementary Notes

A simple criterion to design optimal non-pharmaceutical interventions for mitigating epidemic outbreaks

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S1. Characterization of the optimal intervention in the Susceptible-Infected-Removed model

The model is given by

$$\begin{aligned}\dot{S} &= -(1-u)\beta SI \\ \dot{I} &= (1-u)\beta SI - \gamma I, \\ \dot{R} &= \gamma I\end{aligned}$$

where the parameters $\beta > 0$, $\gamma > 0$ are assumed constant. Since the total population $N = S + I + R$ remains constant all the time, the model can be reduced to that of a second order system using only the states (S, I) . The maximal (acceptable) value of I is $I_{\max}$ and the maximal achievable value of the control is $u_{\max}$. So the state has to belong to the following feasible sets

$$\begin{aligned}\mathcal{X}_F &= \{(S, I) \in \mathbb{R}^2 \mid 0 \leq S \leq 1, 0 \leq I \leq I_{\max}\} \\ \mathcal{U}_F &= \{u \in \mathbb{R} \mid 0 \leq u \leq u_{\max} < 1\}.\end{aligned}$$

Sometimes it will be useful to write the differential equation in a compact form as

$$\begin{aligned}\dot{x} &= f(x) + g(x)w, \quad w = 1 - u \\ \begin{bmatrix} \dot{S} \\ \dot{I} \end{bmatrix} &= \begin{bmatrix} 0 \\ -\gamma I \end{bmatrix} + \beta SI \begin{bmatrix} -1 \\ 1 \end{bmatrix} w.\end{aligned}$$

The trajectory starting at the initial point $x_0 = (S_0, I_0)$ and subject to the control $u : \mathbb{R} \rightarrow \mathcal{U}_F$ is denoted by $\phi(t, x_0, u(\cdot))$.

Let us define the function

$$\Phi_{R_\alpha}(S) = \begin{cases} I_{\max} & \text{if } S < R_\alpha^{-1} \\ I_{\max} + R_\alpha^{-1} (\ln(RS) + 1 - R_\alpha S) & \text{otherwise} \end{cases}$$

with $R_\alpha \in \{R_c, R_0\}$. The optimal control problem consists in finding the control strategy u such that, starting from the initial point (S_0, I_0) , the target set

$$\mathcal{T} = \{(S, I) \in \mathbb{R}_{\geq 0}^2 \mid I \leq \Phi_{R_0}(S)\}$$

is reached in the minimal time with the state restriction $I(t) \leq I_{\max}$ satisfied for all time. Note that this set is positively invariant without control ($u = 0$), and that every trajectory that starts in this set satisfies the restriction $I(t) \leq I_{\max}$ for all $t \geq 0$ (see Fig. S1).

Now let us define the reachable set for an initial state x_0 as the set of points that can be reached from the initial point x_0 with feasible control, i.e.,

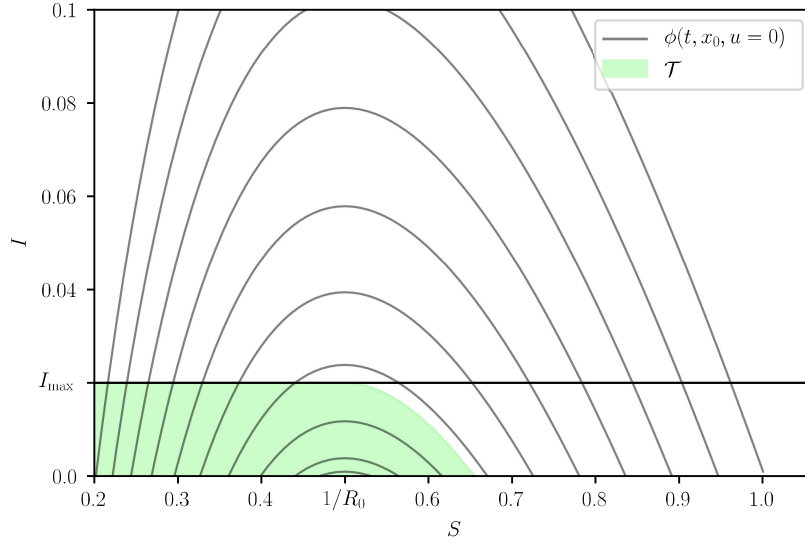
$$\mathcal{R}(x_0) = \{x \in \mathbb{R}_{\geq 0}^2 \mid x = \phi(t, x_0, u(\cdot)) \text{ for some finite } t \geq 0 \text{ and admissible } u\}.$$

Also, we define the controllable set of the target set $\mathcal{T}$ as the set of points from which some point in the target $\mathcal{T}$ can be reached with a feasible control, i.e.,

$$\mathcal{C}(\mathcal{T}) = \{x \in \mathbb{R}_{\geq 0}^2 \mid \exists x_f \in \mathcal{T}, x_f = \phi(t, x, u(\cdot)) \text{ for some finite } t \geq 0 \text{ and admissible } u\}.$$

The set $\mathcal{C}(\mathcal{T})$ can be equivalently described as $\mathcal{R}(\mathcal{T})$ for the system

$$\dot{x} = -f(x) - g(x)(1-u),$$



Supplementary Figure S1 | The set $\mathcal{T}$ is the largest positive invariant set satisfying $I \leq I_{\max}$. The figure was generated with $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

i.e., the set of points that can be reached from the set $\mathcal{T}$ for the dynamics with backward time. Now, the optimal control problem has a solution if and only if

$$\mathcal{R}(x_0) \cap \mathcal{C}(\mathcal{T}) \cap \mathcal{X}_F \neq \emptyset.$$

Since the points of the form $(S, I) = (S, 0)$ are equilibria for every control value, $\mathcal{R}((S, 0)) = (S, 0)$, we exclude them from the initial conditions for which there is a solution (except if the equilibrium is already in the target set). Now, since $\dot{S} < 0$ for $S > 0$, $I > 0$,

$$\mathcal{R}(x_0) \cap \mathcal{C}(\mathcal{T}) \neq \emptyset$$

for every initial condition (except for initial conditions of the form $(S, 0)$). It is obvious that, for the problem to be feasible, the initial state has to be in the feasible set $\mathcal{X}_F$, i.e.,

$$\mathcal{R}(x_0) \cap \mathcal{X}_F \neq \emptyset.$$

S1.1 Calculation of the orbits

Although it does not seem to be possible to find the trajectories of the system explicitly, it is easy to find its orbits. For this we write (we exclude the points for which $I = 0$ since they are equilibria)

$$\begin{aligned} \frac{dI}{dS} &= \frac{\dot{I}}{\dot{S}} = \frac{(1-u)\beta SI - \gamma I}{-(1-u)\beta SI} = \frac{(1-u)\beta S - \gamma}{-(1-u)\beta S} \\ &= \frac{1}{(1-u)R_0 S} - 1 \end{aligned}$$

which is a separable differential equation (DE). Assuming that u is constant and integrating, we obtain

$$I - I_0 = \frac{1}{(1-u)R_0} \ln\left(\frac{S}{S_0}\right) - (S - S_0). \quad (\text{S1})$$

An interesting rewriting of (S1) is

$$I(t) + S(t) - \frac{1}{(1-u)R_0} \ln(S(t)) = I_0 + S_0 - \frac{1}{(1-u)R_0} \ln(S_0) .$$

This means that the quantity $I(t) + S(t) - \frac{1}{(1-u)R_0} \ln(S(t))$ remains constant along the trajectory. Note that this constant depends on the control value used. The above equation is well-known for the SIR model (see, e.g., [1]).

Given an initial condition (S_0, I_0) this expression gives, for any $0 < S < S_0$ the (unique) value of I that is reached in future time¹. Thus there exists a function $I(S; (S_0, I_0))$ that gives the value of I as a function of S and the initial condition. Moreover, from the first equation in the DE we obtain

$$\frac{dS}{(1-u)\beta SI} = -dt$$

and, if we take the expression $I(S; (S_0, I_0))$, we obtain a separable DE that can be integrated,

$$T(S; S_0, I_0) = - \int_{S_0}^S \frac{dS}{(1-u)\beta SI(S; (S_0, I_0))} = - \frac{1}{(1-u)\beta} \int_{S_0}^S \frac{dS}{S \left(I_0 + \frac{1}{(1-u)R_0} \ln\left(\frac{S}{S_0}\right) - (S - S_0) \right)} ,$$

and that gives the time to reach the point $(S, I(S))$ from the initial point (S_0, I_0) with the (constant) control u . Although it does not seem possible to give an explicit expression for this integral, it is clear that S parametrizes uniquely the solutions (since it is monotone).

S1.2 The number of infected people

If we apply a constant control $0 \leq u \leq u_{\max}$ the infection will eventually die out, i.e., the value $I(\infty) = 0$ will be reached asymptotically (otherwise $R(t)$ would continue growing, which is impossible). We can therefore compute $S(\infty)$ implicitly from (S1) as

$$I(\infty) - I_0 = \frac{1}{(1-u)R_0} \ln\left(\frac{S(\infty)}{S_0}\right) - (S(\infty) - S_0)$$

or, equivalently, as

$$\frac{1}{(1-u)R_0} \ln(S(\infty)) - S(\infty) = \frac{1}{(1-u)R_0} \ln(S_0) - S_0 - I_0 .$$

Note that the final value of S depends on the initial values, but also on the control used.

If we assume that the model is normalized, and the initial value is $S_0 = 1$ and $I_0 \approx 0$, then

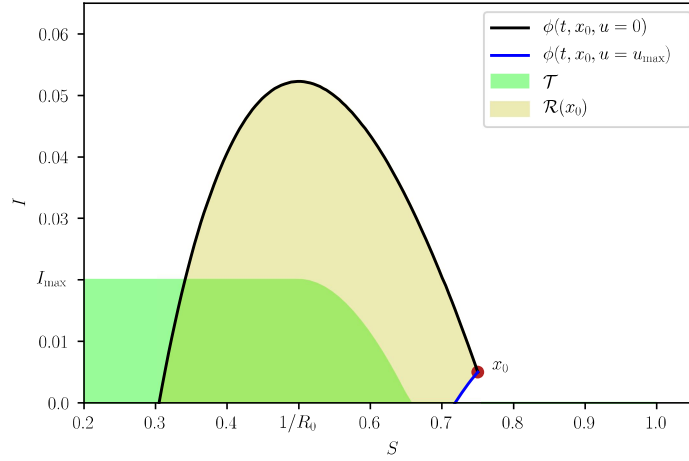
$$S(\infty) - \frac{1}{(1-u)R_0} \ln(S(\infty)) = 1 .$$

Note that, if $u \rightarrow 1^-$, then $S(\infty) \rightarrow 1^-$. So, the larger the value of u , the larger the value of $S(\infty)$.

S1.3 Reachable set from (S_0, I_0)

At each point in the state space, the directions in which the vector field points for different values of the control are given by $F_u(x) = f(x) + g(x)(1-u)$. The extreme values are given by $F_0(x) = f(x) + g(x)$

¹If we select $S > S_0$ the obtained value of I is reached in a past time ($t < 0$).



Supplementary Figure S2 | Phase Plane with maximal and minimal orbits bounding the reachable set $\mathcal{R}(x_0)$. Max corresponds to the trajectory $\phi(t, x_0, u = 0)$ while Min to $\phi(t, x_0, u = u_{\max})$. The figure was generated with $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

and $F_{u_{\max}}(x) = f(x) + g(x)(1 - u_{\max})$,

$$F_0(x) = \begin{bmatrix} 0 \\ -\gamma I \end{bmatrix} + \beta SI \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$F_{u_{\max}}(x) = \begin{bmatrix} 0 \\ -\gamma I \end{bmatrix} + \beta SI \begin{bmatrix} -1 \\ 1 \end{bmatrix} (1 - u_{\max}) .$$

In the phase plane (S, I) both point to the “left”, since the first component (in the direction of S) is always negative (recall that $SI > 0$). Since for the second components of the vector fields we have

$$-\gamma I + \beta SI > -\gamma I + \beta SI(1 - u_{\max}) ,$$

it follows that F_0 is “above” $F_{u_{\max}}$. Therefore, the reachable set $\mathcal{R}(x_0)$ is bounded by the two trajectories $\phi(t, x_0, u = 0)$ and $\phi(t, x_0, u_{\max})$, see Fig. S2. These two bounding orbits can be easily calculated using Eq. (S1).

S1.4 Comparing the cost of two different trajectories

In order to be able to find the orbit (trajectory) solving the optimal control problem, it is necessary to be able to compare the cost of two different trajectories that start at the same initial point and end at the same final point. Consider two orbits $\omega_i(x_0, x_f, u_i)$, $i = 1, 2$, joining the (same) points x_0 and x_f using two different control actions, u_1 and u_2 , respectively. The cost (*i.e.* time) going through ω_i is

$$J(u_i) = \int_0^{T_i} dt$$

along the trajectory. Given two such orbits, we want to compare both costs. This can be done, for example, by subtracting them, *i.e.*, if

$$J(u_1) - J(u_2) < 0$$

then the cost of ω_1 is lower than that of ω_2 .

The cost $J(u_i)$ can be calculated as a line integral along the trajectory. We can see this in the following

manner. Calculate

$$\begin{aligned}\Delta(f(x), g(x)) &= -\det[f(x), g(x)] \\ &= -(f_1(x)g_2(x) - f_2(x)g_1(x)).\end{aligned}$$

Now, by properties of the determinant this is also the same as

$$\begin{aligned}\Delta(f(x) + g(x), g(x)) &= -\det[\dot{x}, g(x)] \\ &= \dot{x}_2g_1(x) - \dot{x}_1g_2(x).\end{aligned}$$

Therefore,

$$\begin{aligned}J(u_i) &= \int_0^{T_i} dt = \int_0^{T_i} \frac{\dot{x}_2g_1(x) - \dot{x}_1g_2(x)}{\Delta(x)} dt \\ &= \int_{x_0}^{x_f} \left(\frac{g_1(x)}{\Delta(x)} dx_2 - \frac{g_2(x)}{\Delta(x)} dx_1 \right),\end{aligned}$$

which is a line integral along the orbit ω_i . Since the two paths have the same initial and final points, they form a closed curve, and calculating the line integral along the closed curve followed in the counterclockwise direction we obtain the difference of the costs, i.e.

$$J(u_1) - J(u_2) = \oint_{\Gamma} \left(\frac{g_1(x)}{\Delta(x)} dx_2 - \frac{g_2(x)}{\Delta(x)} dx_1 \right)$$

where Γ is the closed path of the two orbits followed in the counterclockwise direction. For this we have to assume that: (i) the two paths (orbits) do not intersect at any points except the initial and final ones, and (ii) that $\Delta \neq 0$. Item (i) is without loss of generality: If the orbits do intersect, we can simply use the intersection point as a new final point. Item (ii) is satisfied whenever $S \neq 0$ and $I \neq 0$ (cf. the explicit expression below).

Using Green's theorem, the line integral can be calculated using a surface integral:

$$\oint_{\Gamma} (u(x, y)dy + v(x, y)dx) = \int \int_{\mathcal{R}} \left(\frac{\partial u}{\partial x}(x, y) - \frac{\partial v}{\partial y}(x, y) \right) dx dy = \int \int_{\mathcal{R}} w(x, y) dx dy,$$

where $\mathcal{R}$ is the region enclosed by the closed curve Γ . For our problem this becomes

$$\begin{aligned}J(u_1) - J(u_2) &= \int \int_{\mathcal{R}} w(x_1, x_2) dx_1 dx_2 \\ w(x_1, x_2) &= \frac{\partial}{\partial x_1} \left(\frac{g_1(x)}{\Delta(x)} \right) + \frac{\partial}{\partial x_2} \left(\frac{g_2(x)}{\Delta(x)} \right).\end{aligned}$$

In our case,

$$\begin{aligned}\Delta(x) &= -(f_1(x)g_2(x) - f_2(x)g_1(x)) \\ &= \gamma\beta SI^2 \\ w(x_1, x_2) &= \frac{\partial}{\partial S} \left(\frac{-\beta SI}{\gamma\beta SI^2} \right) + \frac{\partial}{\partial I} \left(\frac{\beta SI}{\gamma\beta SI^2} \right) \\ &= \frac{\partial}{\partial I} \left(\frac{1}{\gamma I} \right) = -\frac{1}{\gamma I^2} < 0.\end{aligned}$$

We see that $w < 0$ everywhere, and therefore the integral is always negative, implying that the “upper” or-

bit has a lower cost than the “lower” orbit (in the closed path traversed in the counterclockwise direction). This observation allows us to find the optimal orbit by comparing it with others.

S1.5 Optimal orbits

From the previous results, the “upper” trajectory is the one with no control ($u = 0$) and, in terms of the cost alone, this trajectory is better than any other one joining the same two points. However, such control may be inadmissible, since the corresponding I can go over $I_{\max}$ at some periods of time.

The computation of the optimal control can be approached in two ways:

- Fix the initial condition x_0 , find its optimal orbit and then its associated optimal control.
- Study the optimal control problem for all possible initial conditions.

Although the second approach is obviously better, it is more difficult, so we will start with the first approach. In fact, both approaches should lead to the same conclusions.

Now we can divide the study of the optimal orbit in several cases.

S1.5.1 Unfeasible trajectories

This is the case if $I_0 > I_{\max}$.

S1.5.2 Trivial trajectories

This is the case in which $I_0 \leq \Phi_{R_0}(S_0)$. That is, the case in which we start in the target set.

S1.5.3 Bang-bang trajectories

If $x_0 \notin \mathcal{T}$, it is necessary to apply some control to maintain I below the maximal value $I_{\max}$. Moreover, admissible trajectories necessarily cross the boundary of $\mathcal{T}$ at $S \geq 1/R_0$, that is, they enter $\mathcal{T}$ at

$$\partial\mathcal{T}_1 = \{(S, I) \mid I = \Phi_{R_0}(S), S \geq 1/R_0\} .$$

In order to find the optimal control that steers an initial state x_0 to $x_f \in \partial\mathcal{T}_1$, consider the change of coordinates

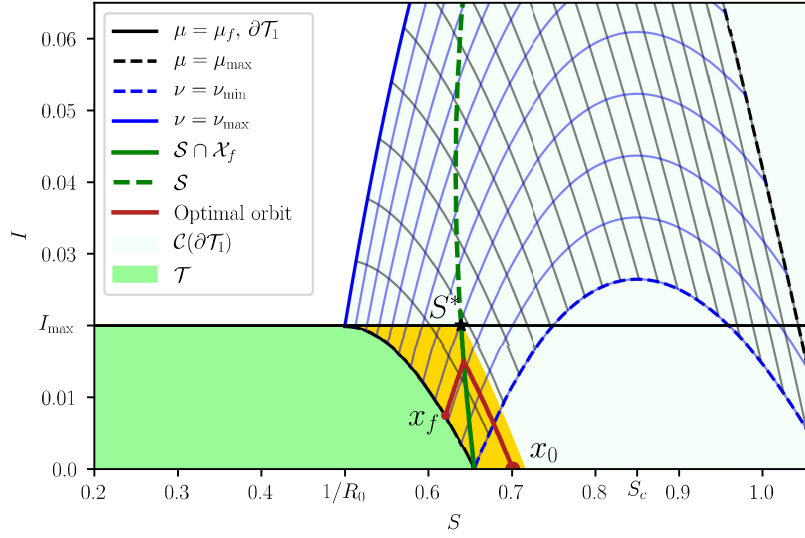
$$\begin{aligned} \mu(S, I) &= I - \frac{1}{R_0} \ln(S) + S \\ \nu(S, I) &= I - \frac{1}{R_c} \ln(S) + S \end{aligned}$$

with inverse

$$S(\mu, \nu) = \exp\left(\frac{1 - u_{\max}}{u_{\max}} R_0(\mu - \nu)\right) \quad (\text{S2a})$$

$$I(\mu, \nu) = \frac{1}{u_{\max}} \mu - \frac{1 - u_{\max}}{u_{\max}} \nu - \exp\left(\frac{1 - u_{\max}}{u_{\max}} R_0(\mu - \nu)\right) . \quad (\text{S2b})$$

Note that, given $I = 0$, we can uniquely map μ to $S \geq 1/R_0$, which we will denote by $S = \hat{S}_\mu(\mu)$. Likewise, we can uniquely map ν to $S \geq 1/R_c$, and we will denote it by $S = \hat{S}_\nu(\nu)$.



Supplementary Figure S3 | An optimal bang-bang trajectory that initiates at x_0 with $u = 0$ (red). When the state touches the curve S (green), the control is set to $u = u_{\max}$ until the state reaches $x_f \in \partial\mathcal{T}_1$. The region containing all bang and bang-bang trajectories is depicted in yellow. The figure was generated with $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

In the new coordinates, the dynamic equations are

$$\dot{\mu} = -uI(\mu, \nu) \quad (S3a)$$

$$\dot{\nu} = \frac{u_{\max} - u}{1 - u_{\max}} I(\mu, \nu) \quad (S3b)$$

Note that $\mu = \text{const}$ is an orbit when $u = 0$ and $\nu = \text{const}$ is an orbit when $u = u_{\max}$.

In (μ, ν) -coordinates, the entry point at $\mathcal{T}$ is the segment

$$\partial\mathcal{T}_1 = \{\mu_f\} \times [\nu_{\min}, \nu_{\max}]$$

with

$$\nu_{\min} = -\frac{1}{R_c} \ln(\hat{S}_\mu(\mu_f)) + \hat{S}_\mu(\mu_f), \quad \nu_{\max} = I_{\max} + \frac{\ln(R_0)}{R_c} + \frac{1}{R_0}$$

and

$$\mu_f = I_{\max} + \frac{\ln(R_0) + 1}{R_0}$$

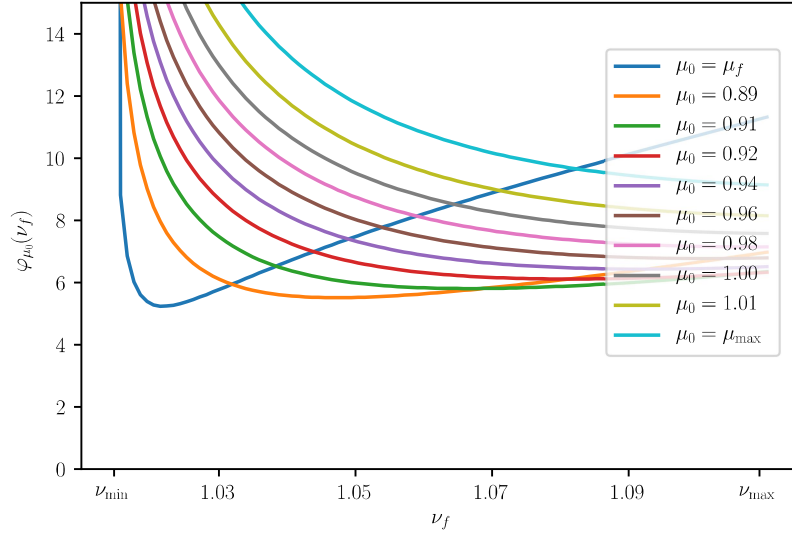
(see Fig. S3).

It follows from Sec. S1.4 that the fastest orbit joining an initial state $(\mu_0, \nu_0) \in \mathcal{C}(\partial\mathcal{T}_1)$ and a final state $(\mu_f, \nu_f) \in \partial\mathcal{T}_1$ is the concatenation of a first piece connecting (ν_0, μ_0) and (ν_f, μ_0) with $u = 0$ and a second piece connecting (ν_f, μ_0) and (ν_f, μ_f) with $u = u_{\max}$. That is, the control is bang-bang. It is easy to verify that this control yields the fastest trajectory, as any other trajectory joining (μ_0, ν_0) and (μ_f, ν_f) is below this one. The transition times can be computed using (S3) as

$$T_0(\nu_f, \mu_0; \nu_0, \mu_0) = \int_{\nu_0}^{\nu_f} \frac{d\nu}{\frac{1}{1-u_{\max}} \mu_0 - \nu - \frac{u_{\max}}{1-u_{\max}} \exp\left(\frac{1-u_{\max}}{u_{\max}} R_0(\mu_0 - \nu)\right)}$$

and

$$T_c(\nu_f, \mu_f; \nu_f, \mu_0) = \int_{\mu_0}^{\mu_f} \frac{d\mu}{-\mu + (1 - u_{\max})\nu_f + u_{\max} \exp\left(\frac{1-u_{\max}}{u_{\max}} R_0(\mu - \nu_f)\right)},$$



Supplementary Figure S4 | The minimal time $\varphi_{\mu}(v_f)$ it takes an initial state $(v_{\min}, \mu_0)$ to reach the target point (v_f, μ_f) . The curves were generated with $R_0 = 2$ and $R_c = 1.18$. Note that the global minimum is well defined (it is unique).

so that the total time is $T(v_f, \mu_f; v_0, \mu_0) = T_0(v_f, \mu_0; v_0, \mu_0) + T_c(v_f, \mu_f; v_f, \mu_0)$.

Note that μ_f is fixed, but $v_f \in [v_{\min}, v_{\max}]$ is free. We will now find the closest entry point by minimizing T over v_f . Set

$$\mu_{\max} = -\frac{1}{R_0} \ln(\hat{S}_v(v_{\min})) + \hat{S}_v(v_{\min}),$$

fix $\mu_0 \in [\mu_f, \mu_{\max}]$ and define the map

$$\begin{aligned} \varphi_{\mu_0} : [v_{\min}, v_{\max}] &\rightarrow \mathbb{R} \\ v_f &\mapsto T(v_f, \mu_f; v_{\min}, \mu_0). \end{aligned}$$

Assumption 1. *Global minima of φ_{μ_0} are unique.*

By Weierstrass Theorem, global minima of φ_{μ_0} always exist. The assumption excludes the highly degenerate case in which the global minimum could occur for more than one value of v_f . Figure S4 shows plots of φ_{μ_0} for various values of μ_0 using the parameters $R_0 = 2$ and $R_c = 1.18$. Note that the global minimum is unique. Also, we have

$$\frac{\partial^2}{\partial v_f^2} \varphi_{\mu_0} = \frac{R_0 R_c}{(R_0 - R_c)^2} \frac{(1 - R_0 S(\mu_0, v_f))}{I^2(\mu_0, v_f)} + \left(\frac{R_c}{R_0 - R_c} \right)^3 \int_{\mu_f}^{\mu_0} \left[2 \frac{(1 - R_0 S(\mu, v_f))^2}{I^3(\mu, v_f)} + R_0^2 \frac{S(\mu, v_f)}{I^2(\mu, v_f)} \right] d\mu.$$

Note that this term is strictly positive for large values of μ_0 , in which case φ_{μ_0} is strictly convex and there exists exactly one global minimum [2]. We now define the function

$$\begin{aligned} v^* : [\mu_f, \mu_{\max}] &\rightarrow [v_{\min}, v_{\max}] \\ \mu_0 &\mapsto \arg \min_{v_f} \varphi_{\mu_0}(v_f). \end{aligned}$$

This function defines a switching curve parameterized by μ_0 . In the original coordinates (S2), the switching curve takes the form

$$\mathcal{S} = \{(S, I) \mid S = S(v^*(\mu_0), \mu_0), I = I(v^*(\mu_0), \mu_0), \mu_0 \in [\mu_f, \mu_{\max}]\}.$$

Let $\bar{I} = \max_{(S,I) \in \mathcal{S}}$. To simplify the exposition, we introduce $\Psi : [0, \bar{I}] \rightarrow [0, 1]$, defined implicitly by

$$(\Psi(I), I) \in \mathcal{S} .$$

We will parameterize $\mathcal{S}$ using I ,

$$\mathcal{S} = \{(S, I) \mid S = \Psi(I), I \in [0, \bar{I}]\} .$$

The trajectories that reach $\mathcal{S}$ above $I_{\max}$ are of course unfeasible, so the class of optimal bang-bang trajectories are only those that pass through $\mathcal{S} \cap \mathcal{X}_F$ (see the yellow region in Fig. S3). For future reference, we will denote by S^* the S coordinate at which $\mathcal{S}$ intersects the line $I = I_{\max}$ and by x_1 the point $(S^*, I_{\max})$.

Summarizing, there are two possible situations:

1. **Bang.** If x_0 belongs to the region delimited by $\partial\mathcal{T}_1$, $I = I_{\max}$ and $\mathcal{S}$, then the optimal control strategy is simply

$$u = \begin{cases} u_{\max} & \text{from } t = 0 \text{ until } x \in \partial\mathcal{T}_1 \\ 0 & \text{when } x \in \mathcal{T} \end{cases} .$$

2. **Bang-Bang.** When x_0 belongs to the region delimited by $\mathcal{S}$, $I = I_{\max}$ and the orbit $\phi(-t, x_1, u_{\max})$, then the optimal control strategy is

$$u = \begin{cases} 0 & \text{from } t = 0 \text{ until } x \in \mathcal{S} \\ u_{\max} & \text{until } x \in \partial\mathcal{T}_1 \\ 0 & \text{when } x \in \mathcal{T} \end{cases} .$$

S1.5.4 Trajectories containing a singular arc

Let us define an initial point $x_0 = (S_0, I_0)$ and the point $x_1 = (S^*, I_{\max})$. We are interested in four trajectories (or orbits):

1. $\phi(t, x_0, u = 0)$, the trajectory without control starting at x_0 . It will be useful to calculate the value $S = \bar{S}$ at which the orbit (first) touches $I_{\max}$. For this we solve (use (S1))

$$I_{\max} = I_0 + \frac{1}{R_0} \ln \left(\frac{S}{S_0} \right) - (S - S_0)$$

for S and obtain two solutions: S_1, S_2 . Define $\bar{S} = \max \{S_1, S_2\}$ as the largest.

Now we calculate the values of $S \leq S_c$ for which it is possible to achieve $\dot{I} \leq 0$ (that is, that it is possible to stop the growth of I). This value can be calculated from

$$\dot{I} = (1 - u_{\max})\beta SI - \gamma I \leq 0$$

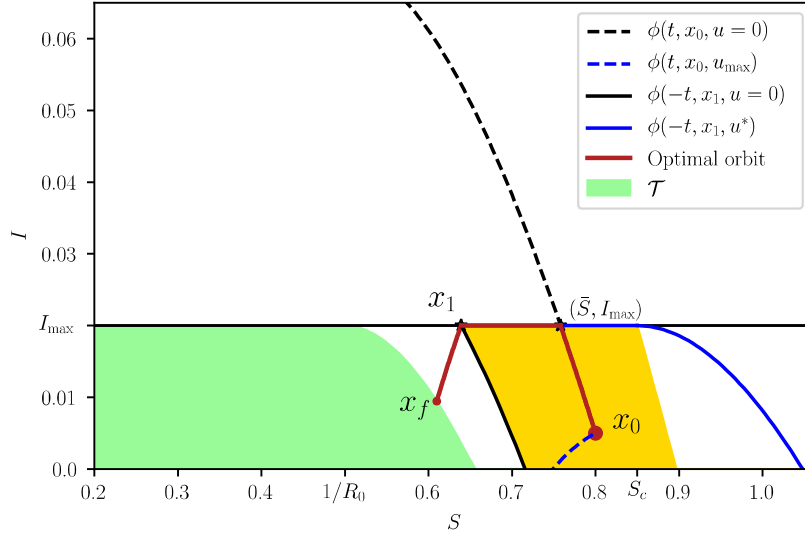
and gives

$$S_c = \min \left\{ \frac{1}{(1 - u_{\max}) R_0}, 1 \right\} .$$

We “saturate” the value of S_c because $S_c > 1$ is not epidemiologically relevant. The control required to achieve the condition $I = I_{\max}$ is the “singular” control

$$u_{\text{sing}} = 1 - \frac{1}{R_0 S} .$$

Note that, if $S > S_c$ at $I = I_{\max}$, it is no longer possible to keep I at $I_{\max}$ because $\dot{I} > 0$.



Supplementary Figure S5 | An optimal trajectory that initiates at x_0 with $u = 0$ (red) and such that $\bar{S} \leq S_c$. When the state touches the curve $I = I_{\max}$, the control is set to $u = u_{\text{sing}}$ until the state reaches x_1 . The control is finally switched to $u = u_{\max}$ until the state reaches the target set. The region containing all bang-singular arc-bang trajectories is depicted in yellow. The figure was generated with $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

If $\bar{S} \geq S_c$, then the optimal control is **bang-singular arc-bang**,

$$u = \begin{cases} 0 & \text{from } t = 0 \text{ until } I = I_{\max} \\ u_{\text{sing}} & \text{until } S = S^* \\ u_{\max} & \text{until } x \in \partial\mathcal{T}_1 \\ 0 & \text{when } x \in \mathcal{T} \end{cases} . \quad (\text{S4})$$

This case is depicted in Fig. S5 for the parameters $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

If $S_c = 1$, then $S_c \geq \bar{S}$ holds trivially and the optimal strategy is again (S4).

2. $\phi(t, x_0, u_{\max})$, the trajectory with maximal control starting at x_0 .
3. $\phi(-t, x_1, u = 0)$, the trajectory without control that passes through x_1 . If x_0 is at the left of this trajectory, the optimal orbit is bang-bang, as shown in the previous section. Optimal trajectories starting at the right have singular arcs.
4. $\phi(-t, x_1, u^*)$, the trajectory with control

$$u^* = \min \{u_{\text{sing}}, u_{\max}\}$$

that passes through x_1 .

The control u^* is such that this trajectory does not violate the restriction $I \leq I_{\max}$. For values of $S \geq S_c$, it is equal to $u_{\max}$, and for $S \leq S_c$ it is the control for the singular arc, i.e., it maintains $I = I_{\max}$ until x_f is reached.

When $\bar{S} > S_c$ then it is necessary to start with the control strategy before reaching the maximal value of $I = I_{\max}$. Otherwise, this limit will be surpassed. However, this is only feasible if, moving backwards from the point $(S_c, I_{\max})$ with the maximal control $u_{\max}$ it is possible to reach a point (S_0, I_c) such that $I_c \geq I_0$. The value of I_c can be calculated from (S1),

$$I_c = (S_c - S_0) + I_{\max} - \frac{1}{(1 - u_{\max}) R_0} \ln \left(\frac{S_c}{S_0} \right) .$$

If $I_c = I_0$, the optimal control is

$$u = \begin{cases} u_{\max} & \text{from } t = 0 \text{ until } I = I_{\max} \\ u_{\text{sing}} & \text{until } S = S^* \\ u_{\max} & \text{until } x \in \partial\mathcal{T}_1 \\ 0 & \text{when } x \in \mathcal{T} \end{cases} .$$

When $I_c > I_0$, the control is **bang-bang-singular arc-bang**,

$$u = \begin{cases} 0 & \text{from } t = 0 \text{ until } S = S_s \\ u_{\max} & \text{from } S = S_s \text{ until } I = I_{\max} \\ u_{\text{sing}} & \text{until } S = S^* \\ u_{\max} & \text{until } x \in \partial\mathcal{T}_1 \\ 0 & \text{when } x \in \mathcal{T} \end{cases} ,$$

where (S_s, I_s) is a switching point. It is characterized as follows: the trajectory $\phi(t, x_0, u = 0)$ intersects the trajectory $\phi(-t, x_1, u_{\max})$ at (S_s, I_s) . Such point can be calculated from (S1) as

$$I_s - I_0 = \frac{1}{R_0} \ln \left(\frac{S_s}{S_0} \right) - (S_s - S_0)$$

$$I_{\max} - I_s = \frac{1}{(1 - u_{\max}) R_0} \ln \left(\frac{S_c}{S_s} \right) - (S_c - S_s) .$$

By substituting the first into the second we get

$$I_s = I_0 + \frac{1}{R_0} \ln \left(\frac{S_s}{S_0} \right) - (S_s - S_0)$$

$$I_{\max} = I_0 + \frac{1}{R_0} \ln \left(\frac{S_s}{S_0} \right) - (S_s - S_0) + \frac{1}{(1 - u_{\max}) R_0} \ln \left(\frac{S_c}{S_s} \right) - (S_c - S_s) .$$

Solving for S_s in the second we arrive at

$$I_s = I_0 + \frac{1}{R_0} \ln \left(\frac{S_s}{S_0} \right) - (S_s - S_0)$$

$$\ln(S_s) = \frac{(1 - u_{\max})}{u_{\max}} \frac{1}{R_0} \left\{ -I_{\max} + I_0 + S_0 - S_c - \frac{1}{R_0} \left(\ln(S_0) - \frac{1}{(1 - u_{\max})} \ln(S_c) \right) \right\} .$$

This case is depicted in Fig. S6 again for the parameters $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

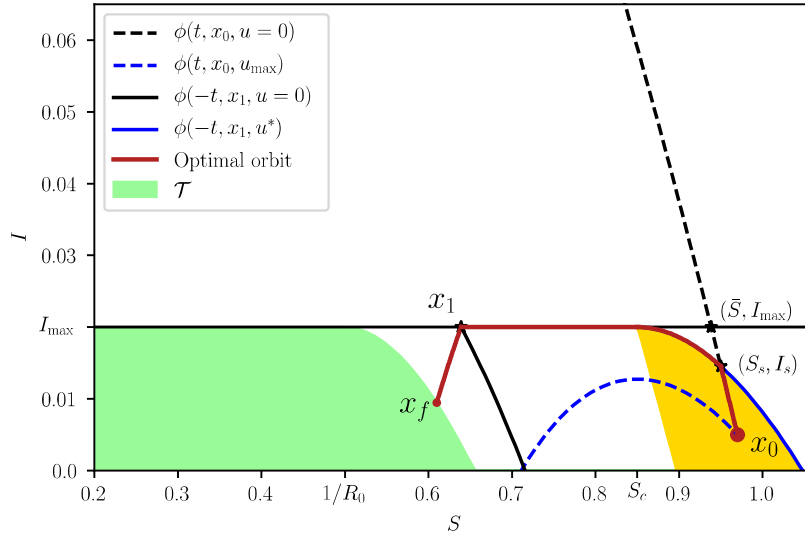
If $I_c < I_0$, then it is not possible to solve the optimal problem, since any strategy will surpass the maximal value $I_{\max}$. This is the case if, e.g., $u_{\max}$ is reduced and R_c increases to 1.27 (see Fig. S7).

S1.6 A feedback control strategy

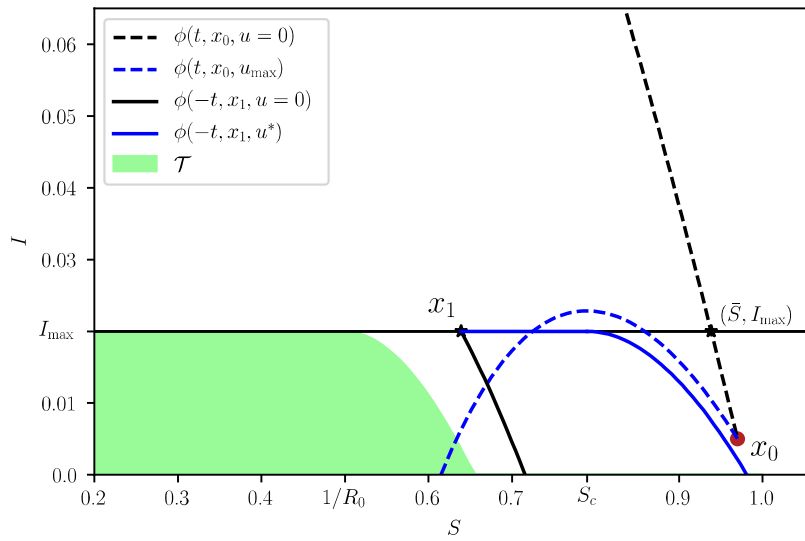
The previous “open loop” strategy can be implemented as a state feedback control. This strategy is rather simple, since there are basically only two switching curves: $\phi(-t, x_1, u^*)$ and $\mathcal{S}$. Another switch takes place when the target region has been attained and the control is switched off, but this happens in a “natural” manner.

The switching curve $\phi(-t, x_1, u^*)$ can be written as

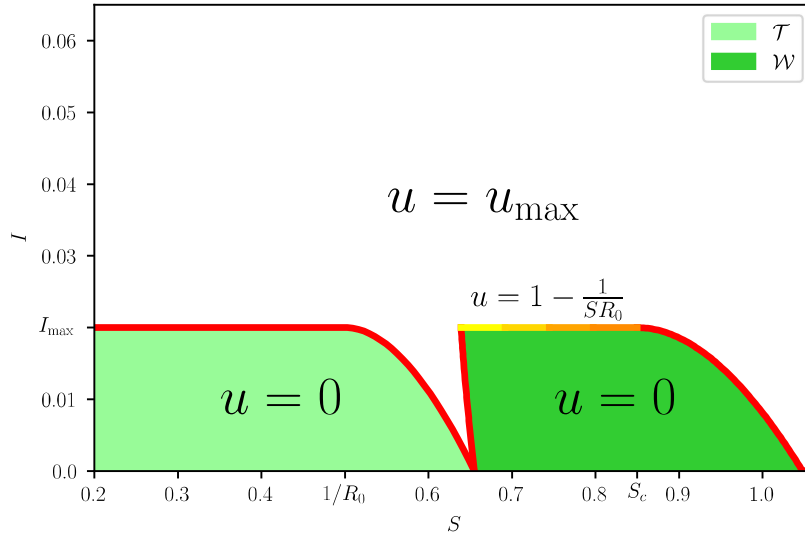
$$I = \Phi_{R_c}(S) , \quad S \geq S^* .$$



Supplementary Figure S6 | An optimal trajectory that initiates at x_0 with $u = 0$ (red) and such that $\bar{S} > S_c$. When the state arrives at (S_s, I_s) , the control is set to $u = u_{\max}$. When the state touches the curve $I = I_{\max}$, the control is set to $u = u_{\text{sing}}$ until the state reaches x_1 . The control is finally switched to $u = u_{\max}$ until the state reaches the target set. The region containing all bang-bang-singular arc-bang trajectories is depicted in yellow. The figure was generated with $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.



Supplementary Figure S7 | For the initial condition x_0 the problem is unfeasible, $I_{\max}$ will be surpassed, no matter which strategy is used. The figure was generated with $R_0 = 2$, $R_c = 1.27$ and $I_{\max} = 0.02$.



Supplementary Figure S8 | The optimal feedback strategy (S5). The target and waiting sets, $\mathcal{T}$ and $\mathcal{W}$, are illustrated for $R_0 = 2$, $R_c = 1.18$ and $I_{\max} = 0.02$.

We can further define the “waiting” set

$$\mathcal{W} = \{(S, I) \mid I < \Phi_{R_c}(S), S > \Psi(I)\} .$$

The optimal control feedback is thus given by

$$u^*(S, I) = \begin{cases} 0 & (S, I) \in \mathcal{T} \cup \mathcal{W} \\ u_{\text{sing}} = 1 - \frac{1}{R_0 S} & \text{if } I = \Phi_{R_c}(S) \text{ and } S^* < S < R_c^{-1} \\ u_{\max} & \text{otherwise} \end{cases} . \quad (\text{S5})$$

Such strategy is summarized in Fig. S8.

Alternatively, we can implement a pure switching control since the “equivalent control”[3] will realize the singular control on the singular arc,

$$u^*(S, I) = \begin{cases} 0 & (S, I) \in \mathcal{T} \cup \mathcal{W} \\ u_{\max} & \text{otherwise} \end{cases} .$$

Note that this control strategy extends the control action beyond the region where the optimal control is feasible.

S2. Necessary and sufficient conditions for the existence of optimal NPIs

Let (S_0, I_0) denote the initial state of the SI model. As shown in Supplementary Note [S1](#), the necessary and sufficient condition for the existence of NPIs is that $I_0 \leq \Phi_{R_c}(S_0)$ where $\Phi_{R_c}(S)$ is the separating curve. To characterize a condition that is independent of the initial state, we consider the limit case of $S_0 = 1$ and $I_0 = 0$. Under this assumption, the necessary and sufficient condition of existence is that $\Phi_{R_c}(1) \geq 0$. In other words, the boundary of existence of NPIs is when the separating curve exactly crosses $I = 0$ at $S = 1$. Substituting $S = 1$ in the separating curve we obtain the condition

$$R_c \leq 1, \quad \text{or} \quad I_{\max} + \frac{1}{R_c} \ln R_c - \left(1 - \frac{1}{R_c}\right) \geq 0,$$

which is precisely the inequality (1) of the Main Text..

S3. Robustness of the optimal intervention

Here we describe the models used to evaluate the robustness of the optimal intervention.

S3.1 Robustness to the presence of demography and an incubation period.

To evaluate the robustness of the optimal intervention to the presence of an incubation period of the disease, we considered the SEIR dynamics

$$\begin{aligned}\dot{S} &= -(1-u)\beta SI + \mu - \mu S, \\ \dot{E} &= (1-u)\beta SI - \lambda E, \\ \dot{I} &= \lambda E - \gamma I, \\ \dot{R} &= \gamma I.\end{aligned}\tag{S6}$$

Above, $E(t)$ denotes the fraction of individuals in the population exposed to the disease, but which are not yet infectious, at time t . The parameter $1/\lambda \geq 0$ denotes the *incubation period* of the disease in units of days. The parameter $\mu \geq 0$ denotes the *recruitment rate* in units of days^{-1} . For the result of our paper, we choose $\mu = 1/(365 \cdot 75)$ corresponding to a life expectancy of 75 years. For this model, the intervention we apply is $u(t) = u^*(S(t), I(t))$ with $u^*(S, I)$ as in Eq. (S5).

S3.2 Robustness to the presence of hidden infected individuals.

To evaluate the robustness of the optimal intervention to hidden infected individuals, consider that that infections can be symptomatic or asymptomatic. We assume that all asymptomatic infections do not require hospital care, and hence remain undetected by the epidemic surveillance system. To model this scenario, we consider the dynamics

$$\begin{aligned}\dot{S} &= -(1-u)\beta S(I_a + I_s) + \mu - \mu S, \\ \dot{E} &= (1-u)\beta SI - \lambda E, \\ \dot{I}_s &= p \lambda E - \gamma I_s, \\ \dot{I}_a &= (1-p) \lambda E - \gamma I_a, \\ \dot{R} &= \gamma(I_s + I_a).\end{aligned}\tag{S7}$$

Above, I_s denotes the fraction of symptomatic infections and I_a the fraction of asymptomatic ones. The model assumes that a fraction $p \in [0, 1]$ of exposed individuals result in symptomatic infections, and the rest $(1-p)$ in asymptomatic ones. We assume that infectious period $1/\gamma$ is the same for both symptomatic and asymptomatic individuals. For the results of our paper, we choose $\lambda = 1/7$. Since we assume that only symptomatic individuals end up requiring hospital care, we consider that the objective is to keep $I_s(t) \leq I_{\max}$ only. The control applied is $u(t) = u^*(S(t), I_s(t))$ where $u^*(S, I)$ is given by Eq. (S5).

S4. Application to the COVID-19 pandemic

S4.1 Estimate for the fraction of infected individuals requiring intensive care.

For COVID-19 pandemic by the SARS-CoV-2 virus, we estimated the fraction f of infected individuals requiring intensive-care under the following assumptions:

1. Current estimates for the fraction $p \in [0, 1]$ of infected individuals that are symptomatic show a large variability [4], ranging from a 20/100 in a report of the World Health Organization, to 96/100 in a study of 328 adults in Shanghai[5]. We take the nominal value of $p_0 = 60/100$.
2. Following Kremer et al.[6], we assume that from the individuals that are symptomatic, a fraction 15/100 develop severe symptoms.
3. Finally, following Li et al. [7], from the individuals that develops severe symptoms, we assume that the fraction 28/100 will require intensive care.

Under the above assumptions, the fraction of infected individuals requiring intensive care has a nominal value

$$f = \frac{60}{100} \frac{15}{100} \frac{28}{100} = \frac{63}{2500} = 0.0252.$$

S4.2 Data used in our analysis.

Supplementary Fig. S9 shows the data used for our analysis. Data was collected using the following methodology:

1. **Number of intensive care beds in each city.** This was obtained from official statements when possible (e.g., the Massachusetts Department of Public Health for Boston). In other cases, this number was obtained from public statements of authorities of each city. A complete list of the references appears in the Supplementary Fig. S9.
2. **Population in each city.** Data was obtained from Wikipedia.
3. **Reduction of mobility in each city.** This was obtained from Google Community Mobility Reports <https://www.google.com/covid19/mobility/>. For our analysis, we considered three categories of mobility: retail & recreation, transit stations, and workplaces. To estimate an overall mobility reduction, we averaged the mobility reduction in these three categories from March 19 to April 30. Data was accessed on May 7, 2020.
4. **Basic reproduction number.** We estimated this quantity from the value of the effective time-varying reproduction number R_t at the start of the pandemic around March 8, 2020. These estimates were obtained from the website <https://epiforecasts.io/covid/>.

COVID-19 (by City)

City	Number of UCI beds	Total population (in millions)	R0 (unmitigated)	R0min (unmitigated)	R0max (unmitigated)	Date of start school closing	Date of start workplace closing	Date of first confirmed case	Average mobility reduction (R)	Average mobility reduction (transit stations)	Average mobility reduction (workplaces)	References
Mexico City	2350	8.855	1.9	1.7	2.4	23/03/2020	26/03/2020	29/02/2020	0.61	0.82	0.49	Declaration of City Mayor to El Universal (April 26) https://www.eluniversal.com.mx/metropoli/com.mx/anuncia-cdmx-500-camias-mas-para-cuidados-intensivos
New York	3673	8.399	2.4	2	3	05/03/2020	19/03/2020	21/01/2020	0.58	0.89	0.58	As in April 12? https://projects.thecitynyc.org/2020_03_covid-19-tracker/
Boston	1600	0.694	2.4	2	3	05/03/2020	19/03/2020	21/01/2020	0.64	0.81	0.6	Massachusetts Department of Public Health COVID-19 Dashboard - Tuesday, May 05, 2020 https://www.mass.gov/doc/covid-19-dashboards-may-5-2020/download Mobility reduction in Suffolk County
Dehli (India)	1973	21.75	1.9	1.4	2.5	04/03/2020	16/03/2020	30/01/2020	0.89	0.82	0.81	April 20, Princeton Center For Disease Dynamics, Economics & Policy https://cddp.org/wp-content/uploads/2020/04/State-wise-estimates-of-current-beds-and-ventilators_20Apr2020.pdf
Lombardy (Italy)	1330	10.06	2.2	1.8	2.4	23/02/2020	22/02/2020	31/01/2020	0.71	0.71	0.57	March 13, Critical Care Utilization for the COVID-19 Outbreak in Lombardy, Italy. https://jamanetwork.com/journals/jama/fullarticle/2763188 https://ec.europa.eu/icc/en/news/forecasting-covid-19-impact-icu-bed-demand-lombardy
Berlin	2267	3.769	2.2	1.8	2.5	26/02/2020	22/02/2020	28/01/2020	0.7	0.5	0.5	April 2, berlinreporter.com/2020/04/02/coronavirus-crisis-berlin-intensiv-care-capacities-increased-substantially/ https://www.tagesspiegel.de/berlin/mueller-zu-besuch-in-den-charite-wie-sich-berlins-grossstes-krankenhaus-auf-den-corona-ernstfall-vorbereitet/25714372.html
Tokyo	848	9.273	1.3	1.1	2.2	02/03/2020	25/02/2020	15/01/2020	0.54	0.57	0.5	April 26 https://www.japantimes.co.jp/news/2020/04/26/national/science-health/japan-icu-bed-shortage-most-prefectures-coronavirus-peak/#:XfmwYc2ZQCM
Madrid	1750	6.642	3.11	2.86	3.37	09/03/2020	09/03/2020	01/02/2020	0.86	0.79	0.73	April 3 https://as.com/diarios/2020/04/03/actualidad/1585905064_448016.html
Lima	517	8.575	2.2	1.7	2.8	12/03/2020	16/03/2020	07/03/2020	0.75	0.8	0.54	April 29 https://www.gob.pe/institucion/minsa/noticias/142142-peru-impone-822-camias-uci-para-la-atencion-de-pacientes-covid-19
British Columbia (Canada)	2398	5.071	2.2	1.8	2.7	16/03/2020	18/03/2020	26/01/2020	0.43	0.56	0.52	March 20 https://www.cbc.ca/news/canada/british-columbia/bc-hospital-beds-covid-19-1.5505356
Athens	556	0.664	1.75	1.4	2.3	05/03/2020	12/03/2020	27/02/2020	0.65	0.7	0.5	April 8, https://publichealth.uga.edu/athens-covid-19-hospitalizations-new-report/ Mobility data of Decentralized Administration of Thessaly and Central Greece
Buenos Aires	2200	2.89	2.1	1.6	2.8	16/03/2020	19/03/2020	04/03/2020	0.82	0.7	0.66	March 26 https://www.lanacion.com.ar/sociedad/coronavirus-salud-preparadas-terapias-intensivas-de-pais-rid-347722
Vienna	1101	1.89	2.2	1.8	2.6	16/03/2020	12/03/2020	26/02/2020	0.7	0.7	0.7	May 7, Austrian Ministry of Health https://info.gesundheitsministerium.at/dashboard_hosp.html?en
New South Wales (Australia)	874	7.544	2	1.7	2.5	22/03/2020	24/03/2020	25/01/2020	0.43	0.62	0.4	March 19, https://theconversation.com/how-well-avoid-australas-hospitals-being-crippled-by-coronavirus-133920 https://www.health.nsw.gov.au/Infectious/covid-19/communities-of-practice/documents/covid-19-adult-intensive-care-pandemic-response-planning.pdf
Dublin	153	1.3	2.3	1.7	2.7	12/03/2020	12/03/2020	01/03/2020	0.8	0.76	0.73	May 3, https://www.rte.ie/news/2020/05/03/1136299-coronavirus-ireland/
Bogota	1555	7.413	2	1.6	2.6	16/03/2020	24/02/2020	07/03/2020	0.75	0.73	0.7	April 14, https://www.eltiempo.com/colombia/otras-ciudades/camias-en-colombia-para-atender-a-paciente-con-coronavirus-484048

Supplementary Figure S9 | Table with the response of 16 cities during the COVID-19 pandemic.

S5. Related work

For the control of infectious diseases, there is a large body of work using optimal control methods to design interventions, including vaccination and quarantines[8, 9], drug treatments[10], or dispersal of insecticides and education campaigns[11]. The standard tool to solve these optimal control problem is the celebrated Pontryagin’s Maximum Principle[12]. However, note that the Maximum Principle only gives necessary conditions for optimality. The gap between the necessary and sufficient conditions for optimality needs to be closed using additional arguments, often relying on assuming that the control appears multiplying an affine function of the state variables. This assumption is not satisfied in our formulation of optimal NPIs. We emphasize that the optimal interventions obtained from this approach result in *open loop* strategies which only depend on time. By contrast, our analysis gives a feedback optimal strategy that characterizes the optimal action to make according to the actual state of the epidemic. Indeed, our characterization of optimal NPIs does not rely on the Maximum Principle. Instead, the low dimensional of our model allows us to apply Green’s Theorem to compare the cost of two different interventions. The consequence of our approach is that we obtain a feedback or *closed loop* strategy that corrects itself based on the actual state of the epidemic.

The COVID-19 pandemic has stirred much interest on designing non-pharmaceutical interventions. This has led to strategies like interspacing mitigation with brief periods of activity[13]. Optimal control methods have been also applied, for example to minimize the peak of infection[14], minimize the number of infections[15], minimize the economic costs[16], or maximize welfare[17]. Compared to these studies, our analytical characterization of optimal NPIs provides gives us a complete understanding of the optimal decisions that need to be made. For example, no intervention is needed before reaching the separating curve.

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